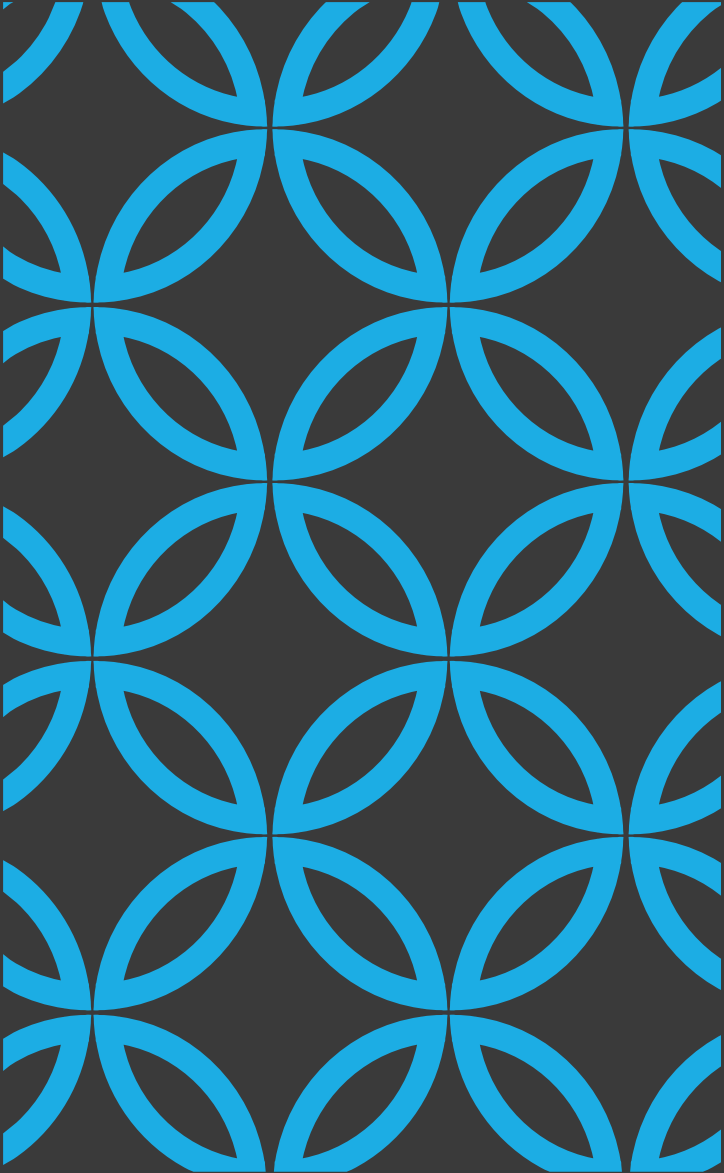




STRATEGIC RETAIL EXPANSION IN EUROPEAN CITIES

A Data-Driven Market Entry Analysis using Principal Component Analysis (PCA)

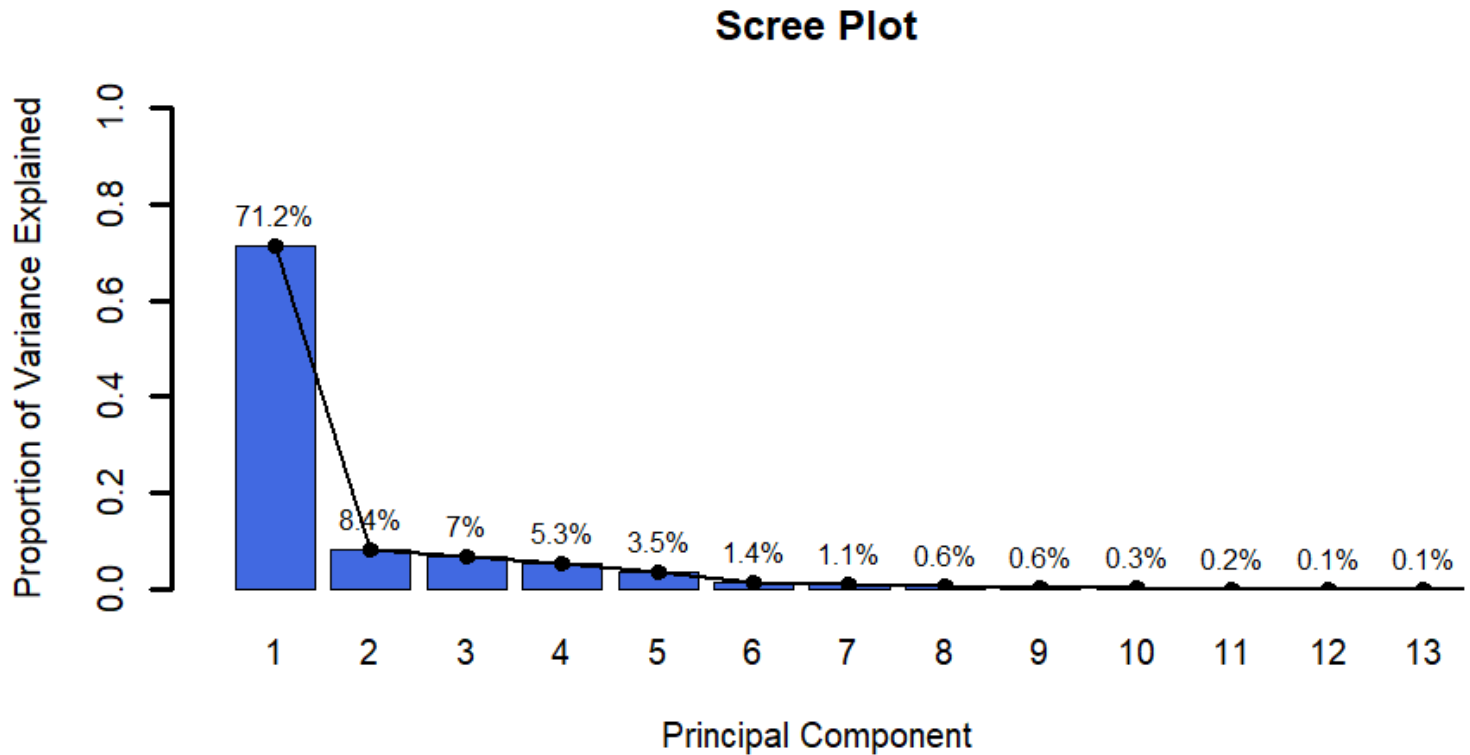


NAVIGATING COMPLEXITY IN MARKET SELECTION

April 2026

- **The Problem:** Evaluating 800+ cities across 27 different metrics creates information overload.
- **The Risk:** Many indicators are strongly correlated (eg Population and Total Jobs), leading to biased rankings if viewed separately.
- **The Objective:** Eliminate correlation bias and simplify the decision-making process by identifying the underlying Market Dimensions.

DIMITRIOS KOSTINIS | PCA MARKET ANALYSIS



FROM 27 INDICATORS TO 2
STRATEGIC DIMENSIONS

- We utilized PCA to create new, uncorrelated variables.
- Our model retains more than 80% of the original information concentrated into 3 components.
- **Result:** A clear, mathematically objective ranking system

A low-angle, upward-looking photograph of several tall, modern skyscrapers in a city. The buildings are made of concrete and glass, with many windows. The sky is blue with some white clouds. A palm tree is visible in the bottom right corner. The overall perspective makes the buildings appear to converge towards the top of the frame.

THE MARKET GIANTS

Characteristics: High Population, Employment and Infrastructure.

Top Performers: Paris, Berlin, Madrid.

Strategic Fit: High volume operations requiring stable consumer bases.

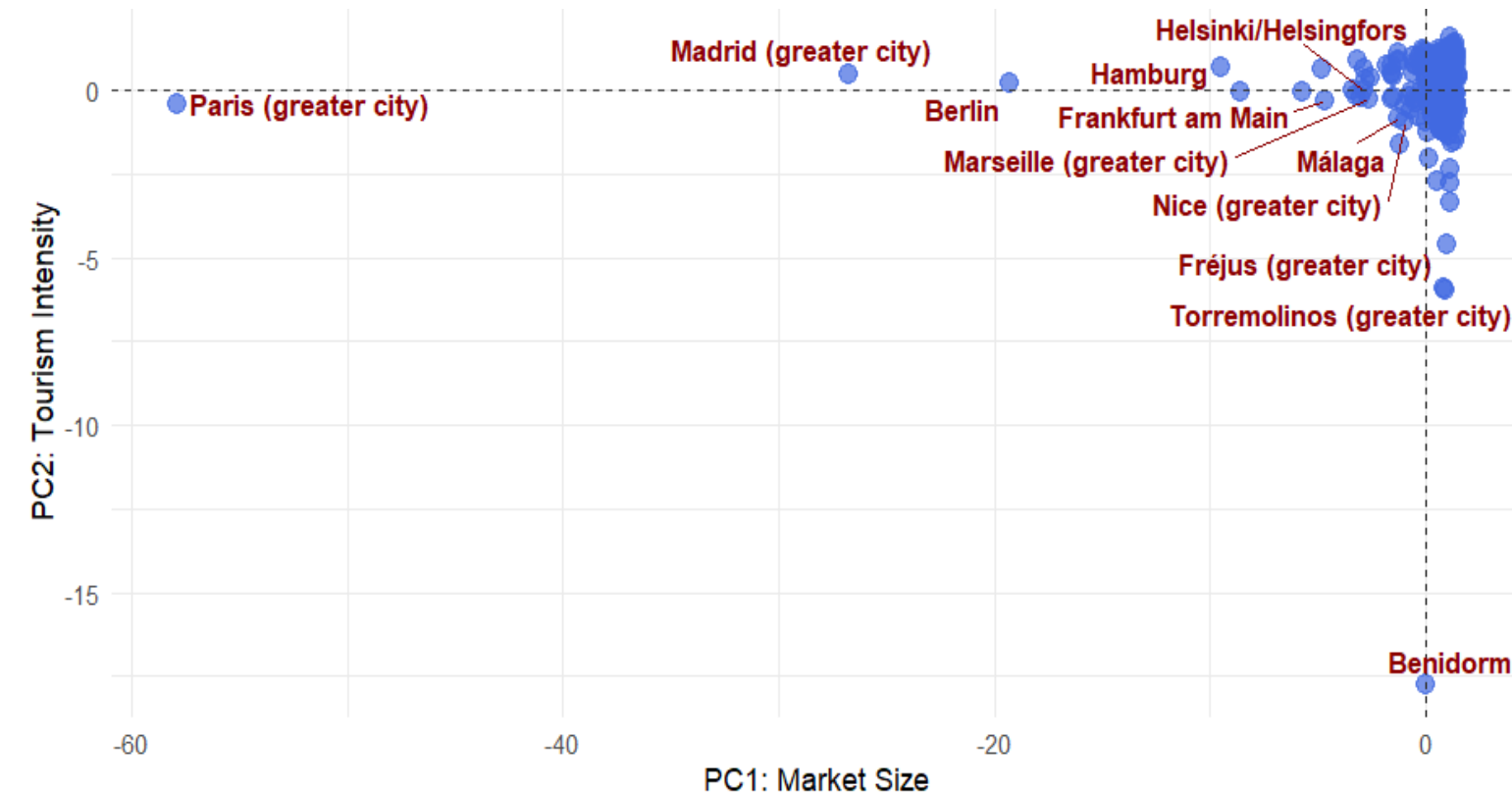


THE TOURIST HUBS

Characteristics: High Tourism Intensity, High available beds to residents proportion.

Top Performers: Benidorm, Nice, Malaga.

Strategic Fit: Luxury brands, concept Hospitality and experiences.



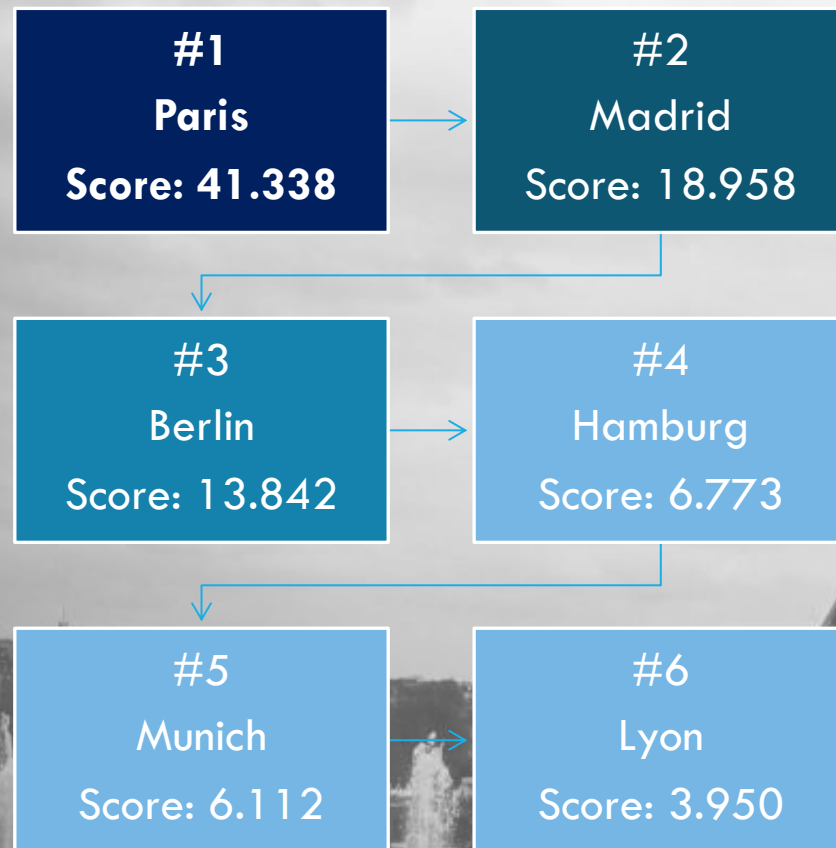
Top Right Quadrant:
Baseline cities; balanced size and tourism.

Top Left Quadrant:
Economic giants.

Bottom Right Quadrant:
Tourist hubs with growth opportunities.

VISUALIZING WHERE THE CITIES
STAND

FINAL RECOMMENDATION



TOP 6 most Attractive cities for retail expansion:

These cities represent the most balanced and attractive environments for retail entry based on our composite index.